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Pareto-Optimal Parametric Design Method for PMSM Applied to eVTOL UAV

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***Abstract*—There is a growing interest in the development of electric propulsion systems for vertical take-off and landing (eVTOL) unmanned aerial vehicles (UAV). These vehicles require electric machines with a high torque-to-mass ratio and high efficiency. Surface-mounted permanent magnet synchronous machines (SM-PMSM) with outer-rotor configurations remain a focus of attention for this application, as they meet the required performance characteristics and, additionally, exhibit mechanical features that facilitate direct coupling with the propeller. The optimal design of an SM-PMSM is a complex task, due to the large number of interacting design variables. Furthermore, the adopted design strategy strongly depends on the key performance indicators (KPI) associated with the target application. In this paper, a parametric framework to identify Pareto-optimal outer-rotor SM-PMSM designs within a predefined design space is presented. The construction of the design plane is based on an analytical model of outer-rotor permanent magnet machines. Finite element analysis (FEA), carried out using FEMM, is applied to the design selected as a validation of the proposed analytical design model. This approach enables the selection of the most suitable machine for a given application from a family of designs. Using specific torque and efficiency as the primary design drivers.**

***Keywords*—PMSM, BLDC, outer-rotor, Pareto-optimal selection, Parametric design, Analytical model.**

I. Introduction

The rapid growth of electric vertical take-off and landing vehicles (eVTOL) is transforming urban air mobility, positioning electric machines as key enabling components. High torque density and high efficiency are essential to meet stringent requirements in terms of mass, range, and reliability [1] [2]. Since eVTOL aircraft must generate sustained vertical thrust during take-off and landing, the specific torque $(\mathrm{Nm/kg})$ of the electric machine becomes a critical factor for minimizing the propulsion system mass [3]. Additionally, high efficiency is necessary to maximize range when operating with batteries, given their limited energy capacity.

Among the available electromagnetic topologies, outer-rotor permanent magnet machines offer significant advantages for eVTOL applications. Their large airgap radius relative to the machine size results in a high torque-to-volume ratio, making them well-suited for direct-drive propulsion systems with large-diameter propellers [4]. Moreover, rotor integration with structural elements enables compact, lightweight designs, thereby improving the power-to-weight ratio and reducing mechanical complexity [3], [5], [6]

Despite these advantages, the design of such machines remains challenging because it requires simultaneously optimizing multiple geometric and electromagnetic parameters subject to thermal, structural, and manufacturing constraints. In this context, systematic parametric design methodologies are essential for efficiently exploring the design space and ensuring reproducibility [7] [8]. Furthermore, multi-objective optimization techniques, such as Pareto-based approaches [9] [10] [11], provide a structured framework for identifying optimal trade-offs.
Unlike prior parametric design approaches focused on field-weakening operation for traction applications, the proposed method explicitly targets direct-drive outer-rotor surface-mounted permanent magnet synchronous machines SM-PMSM operating at nominal speed, using specific torque and efficiency as the primary design drivers.

This paper proposes a parametric design framework for outer-rotor SM-PMSM for eVTOL UAV applications. The methodology includes a parametric design plane that represents the parametric design space in which the optimal machine is searched, and three Pareto-optimal selection methods. The methodology is demonstrated through a case study and validated using finite element method (FEM) simulations.

Although thermal analysis is a decisive topic in motor design, it is not covered in the present work, as it focuses on electromagnetic design.

II. Application to eVTOL UAV

The proposed design methodology aims to identify electric machines that are best suited for vertical take-off and landing (VTOL) applications. In this work, specific torque $(\mathrm{Nm/kg})$ and efficiency are selected as key performance indicators (KPI), as they directly quantify the suitability of a given machine for eVTOL propulsion.

In these aircraft, mass plays a fundamental role, as it directly affects both the required power and the onboard energy storage. Motor efficiency is equally important, since it impacts battery energy requirements and thermal management, ultimately influencing the total vehicle mass. For a given magnetic flux density B, the torque capability of an electric machine is proportional to its physical size and achievable current density, and therefore to its mass.

However, the machine with the highest specific torque within the parametric design plane generally does not exhibit the highest efficiency under identical operating conditions, revealing an inherent trade-off between these two metrics.

The aerodynamic power required by a propeller can be expressed as [12]

$$P_{aero} = \frac{Thrust_{aero}^{3/2}}{FOM\sqrt{2\rho A_{disk}}} \tag{1}$$

Consequently, the driving torque T_m can be written as

$$T_m = \frac{P_m}{\omega_m} = \frac{Thrust_{aero}^{3/2}}{\omega_m FOM\sqrt{2\rho A_{disk}}} \tag{2}$$

On the other hand, the aerodynamic thrust $Thrust_{aero}$ is given by

$$Thrust_{aero} = C_t\,\rho\left(\frac{\omega_m}{2\pi}\right)^2(2R)^4 \tag{3}$$

Where $Thrust_{aero}$ denotes the propeller thrust, FOM is the propeller figure of merit, ρ is the air density, A_{disk} is the propeller disk area, ω_m is the angular speed, C_t is the thrust coefficient, and R is the propeller radius.

By combining the previous expressions and grouping all constants into a single coefficient, the required torque as a function of rotational speed can be expressed as

$$T_m = K\rho R^5\omega_m^2 \tag{4}$$

As can be observed, the resistive torque profile of a propeller increases with the square of the rotational speed. Therefore, for eVTOL applications, the torque capability of the electric machine at nominal rotational speed is more relevant than its low-speed performance. This load characteristic naturally favors maximum torque per ampere (MTPA) operation, whereas operation in the field-weakening region, which reduces the maximum torque, is neither required nor desirable.

III. Magnetic flux Model

For an ideal balanced three-phase symmetrical winding with half-wave symmetry, the resultant air-gap MMF contains predominantly the fundamental and odd space harmonics (even orders cancel). The fundamental space harmonic is responsible for the average (useful) electromagnetic torque, while higher spatial/time harmonics generate parasitic/pulsating torque components (torque ripple) and additional losses [13]. Assuming a uniform magnetization and neglecting higher-order spatial harmonics, the air-gap magnetic flux density can be approximated by its fundamental component, as we can see in [14]:

$$B_{g1}(\theta) = \frac{4B_g}{\pi}\,\sin\!\left(\frac{\alpha_m}{2}\right)\sin(\theta), \tag{5}$$

where B_g is the peak air-gap flux density and $0 < \alpha_m < \pi$ is the electrical magnet span angle.

The magnetic flux crossing the effective area can be expressed as [15]:

$$\Phi_1(\theta) = \Phi_{1,\max}\sin(\theta), \qquad \Phi_{1,\max} = \frac{4B_g}{\pi}\,\sin\!\left(\frac{\alpha_m}{2}\right)A_{\text{eff}}, \tag{6}$$

where A_{eff} denotes the effective magnetic area. The magnet-induced flux linkage is then obtained as

$$\lambda_m(\theta) = N_s\Phi_1(\theta), \tag{7}$$

Being N_s the number of conductors in series per phase.
For the fundamental harmonic responsible for torque production, it yields

$$\lambda_{m1}(\theta) = N_s\left[\frac{4B_g}{\pi}\sin\!\left(\frac{\alpha_m}{2}\right)A_{\text{eff}}\right]\sin(\theta). \tag{8}$$

The induced back-electromotive force (EMF) is obtained from the time derivative of the flux linkage,

$$e(t) = -\frac{d\lambda_m}{dt} = -\frac{d\lambda_m}{d\theta_e}\,\omega_e. \tag{9}$$

Restricting the expression to the fundamental harmonic leads to

$$e_1(t) = -N_s\Phi_{1,\max}\,\omega_e\cos(\theta), \tag{10}$$

which is used in the subsequent parametric design and validation stages.

IV. Cogging Torque

Cogging torque is an undesired effect arising from the non-continuous magnetization of the rotor, combined with the variation of the air-gap reluctance as a function of the rotor mechanical position angle θ_m. As a result, the rotor tends to align at the angular positions θ_m corresponding to minimum magnetic reluctance paths. Therefore, it is highly desirable to keep the cogging torque as low as possible. For this reason, cogging torque expressions are incorporated into the design methodology, allowing the design space to be restricted to machines exhibiting minimal cogging torque for a given pole pair (p) and slot (Q_s) combination. In general, cogging torque is defined as the derivative of the magnetic co-energy with respect to the rotor position when phase current $i_{ph} = 0$ [15]

$$T_{cog(\theta_m)} = -\frac{\delta W(\theta_m)}{\delta\theta_m}.$$

In [16], an analytical model of the cogging torque is developed, whose general form can be expressed as

$$\begin{aligned} T_{cog}(\theta_m) = {} & -\frac{L_{st}B_g^2C_T}{\mu_0\,\pi}(R_{go}^2 - R_{gi}^2) \\ & \times\sum_{n=1}^{\infty}\frac{K_{skw}}{n}\sin\!\left(n\frac{N_L\pi k_{so}}{Q_s}\right)\sin\!\left(nN_L\frac{\alpha_m}{2p}\right) \\ & \times\sin\!\left(nN_L\alpha\frac{1}{2}N_L\alpha_s\right) \end{aligned} \tag{11}$$

$$C_T = \frac{Q_s\, 2p}{N_L} \quad , \quad N_L = LCM(Q_s, 2p) \tag{12}$$

where L_{st} is the stator axial length, C_T is the cogging factor, R_{go} and R_{gi} are the outer and inner air-gap radii, respectively, K_{skw} is the skew factor, and k_{so} denotes the ratio between the slot opening angle and the slot pitch. In this work, skewing is not employed in order to prioritize torque density; therefore, $K_{skw} = 1$ and $\alpha_s = 0$.

A. Cogging Torque Minimization

As indicated in [16], by examining (11), it can be observed that the cogging torque is minimized when

$$\sin\left(nN_L\frac{\alpha_m}{2p}\right) = 0.$$

From this condition, taking the fundamental value of the cogging torque only ($n = 1$), the magnet pitch angle α_m that minimizes the cogging torque can be obtained as

$$\alpha_m = k\frac{2p}{N_L}\pi \qquad (k = 1, 2, 3, \ldots, \frac{N_L}{2p} - 1)$$

In this work, maximizing specific torque is a primary objective. Therefore, the largest possible magnet pitch α_m is selected, corresponding to $k = \frac{N_L}{2p} - 1$, which yields

$$\alpha_m = \frac{N_L - 2p}{N_L}\pi \tag{13}$$

Additionally, cogging torque is also minimized when

$$\sin\left(n\frac{N_L \pi k_{so}}{Q_s}\right)$$

The value of k_{so} that satisfies the above condition is given by

$$k_{so} = \frac{k\, Q_s}{N_L} \qquad (k = 1, 2, 3, \ldots, \frac{N_L}{Q_s}).$$

V. Analytical Design Model

A. Parametrization

Parametric design of electric machines is not a new topic. In [17], the ratio between the maximum magnetic flux density in the airgap (B_g), and the maximum allowable magnetic flux density in the in the stator core ($B_{h_{max}}$) is employed for the calculation of machine parameters, such as the PMSM inductance. In [18], parametric design is applied to interior permanent magnet (IPM) machines, while [19] addresses the parametric design of surface-mounted permanent magnet (SPM) machines. However, in these works, the primary objective is to operate the machine in the field-weakening region for ground vehicle traction applications. In particular, the latter develops a procedure to ensure constant power operation in that region, at the expense of reducing the torque produced by the machine at base speed—defined as the rotational speed prior to field weakening—by a factor of $\frac{1}{\sqrt{2}}$. As a consequence, torque density and efficiency are affected by this constraint and become secondary outcomes of a design process governed by a fixed value of $\cos(\phi)$. In contrast, the objective of this work is to identify machines exhibiting the highest torque density

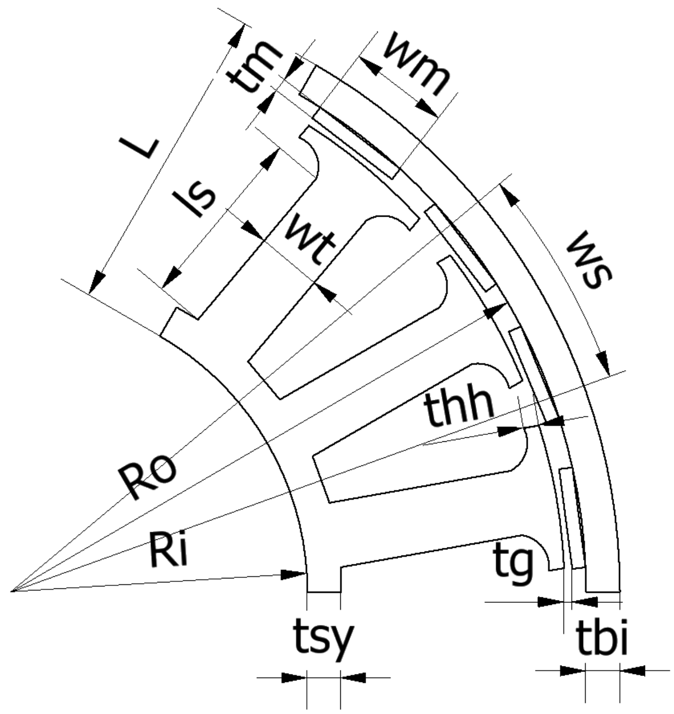


Fig. 1: Design Motor Parameters.

and efficiency, while $\cos(\phi)$ is treated as a secondary outcome that naturally follows from the main design objectives. To this end, a new parametric design plane (l, b) is introduced, as defined in the following. Furthermore, as established in the application definition, the proposed methodology is applied to surface-mounted permanent magnet synchronous machines (SM-PMSM) with outer-rotor topology. Consequently, the selected parametrization is particularly well suited to this machine architecture. In prior works [20] [21], an analytical model was presented, and here it is further developed.
The parameters defining the (l, b) design plane are given by

$$b = \frac{B_{g_1}}{B_{h_{max}}} \quad ; \quad l = \frac{L}{R_o} = \frac{R_o - R_i}{R_o} = 1 - \frac{R_i}{R_o}$$

R_o and R_i are the outer and inner radius of the machine, as are defined in Fig. 1. The quantity $L = R_o - R_i$ represents the radial length of the active electromagnetic section of the machine. An analysis of the selected parameters shows that parameter b governs the relationship between the tooth width (w_l) and the available slot arc (w_s), since it relates the air-gap flux density to the maximum magnetic flux density permitted in the ferromagnetic material. This parameter also influences the air-gap radius R_g, as B_g is strongly dependent on the magnet thickness t_m. In addition, both the rotor back-iron thickness (t_bi) and the stator yoke thickness (t_sy) are closely linked to parameter b. Consequently, this parameter also affects the slot length l_s. Parameter l relates the proportion of the electromagnetically active region of the machine to the outer (total) radius. Therefore, it indicates whether the machine exhibits a ring-like geometry for $l \leq 0.5$ or a more disk-like geometry for $l > 0.5$.

B. Parametric Model

In the following, the general equations developed in the previous sections are applied to an outer-rotor PMSM (OR-PMSM). Starting from Ampere's, the air-gap magnetic flux density is obtained as

$$B_g = k_l \frac{B_r}{1 + \mu_r \frac{t_g}{t_m}} \tag{14}$$

where k_l is the leakage factor, B_r is the magnetic remanence, μ_r is its relative permeability, t_g is the air-gap thickness, and t_m is the magnet thickness. The waveform factor k_b is defined in [14] as

$$k_b = \frac{B_{g_1}}{B_g} \implies B_g = \frac{B_{g_1}}{k_b}. \tag{15}$$

Substituting (5) and (15) into (14) yields

$$B_{g_1} = k_b\, k_l \frac{B_r}{1 + \mu_r \frac{t_g}{t_m}}. \tag{16}$$

The resulting magnet thickness is then obtained as

$$t_m = \frac{\mu_r\, t_g}{\frac{k_b\, k_l\, B_r}{b\, B_{h_{max}}} - 1}. \tag{17}$$

From (8), the effective area A_{eff} is defined as

$$A_{\text{eff}} = k_w L_{st} R_g \frac{\pi}{p},$$

where k_w is the winding factor. Substituting A_{eff} into the flux linkage expression (8), taking its maximum value, and applying the appropriate transformations, the d-axis flux linkage is obtained as

$$\lambda_{\text{d}} = \frac{4 N_s B_g k_w L_{st} R_g}{\sqrt{3}\, p} \sin\left(\frac{\alpha_m}{2}\right). \tag{18}$$

Expressing the previous equation as a function of parameter b yields

$$\lambda_{\text{d}} = \frac{\pi N_s B_{h_{max}} k_w L_{st} R_g}{\sqrt{3}\, p}\, b. \tag{19}$$

For a surface-mounted permanent magnet machine with d and q inductance $L_d = L_q = L$, the electromagnetic torque is given by

$$T_m = \frac{3}{2}\, p\, \lambda_d I_q.$$

Thus,

$$T_m = \frac{\sqrt{3}}{2} \pi N_s B_{h_{max}} k_w L_{st} R_g I_q\, b = k_t\, I_q. \tag{20}$$

Similarly, substituting λ_m into (10) and taking its peak value yields

$$\hat{E}_1 = \frac{\pi N_s B_{h_{max}} k_w L_{st} R_g}{\sqrt{3}}\, b\, \omega_m. \tag{21}$$

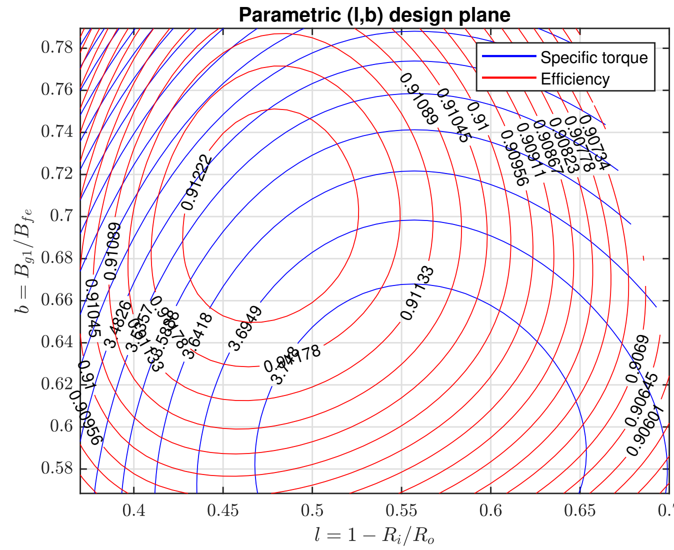


Fig. 2: Parametric design plane.

To reduce back-iron mass, the back-iron is designed to operate at the maximum magnetic flux density prior to saturation, $B_{h_{max}}$, resulting in

$$t_{bi} = b(R_o - 2t_m)\sin\left(\frac{\alpha_m}{2p}\right) = b(R_o - 2t_m)\sin\left(\frac{k_m \pi}{2p}\right). \tag{22}$$

Here, k_m is the magnet coverage factor. The resulting air-gap radius is given by

$$R_g = R_o - \left(t_{bi} + t_m + \frac{t_g}{2}\right). \tag{23}$$

The magnet width is defined from the magnet pitch obtained through cogging torque minimization as

$$w_m = 2(R_o - t_{bi} - t_m)\sin\left(\frac{\alpha_m}{2p}\right). \tag{24}$$

The opening arc of the pole expansion is defined as

$$w_{hh} = 2\pi \frac{R_g}{Q_s}(1 - k_{so}). \tag{25}$$

The stator tooth width is calculated as

$$\begin{cases} w_t = b \cdot w_m, & w_{hh} > w_m, \\ w_t = b \cdot w_{hh}, & w_{hh} \leq w_m, \end{cases} \tag{26}$$

Finally, the slot length is obtained as

$$l_s = l \cdot R_o - (t_{bi} + t_m + t_g + t_{hh} + r_{hh} + \frac{l_s}{2}). \tag{27}$$

Being t_{hh} and r_{hh}, the pole expansion thickness and fillet radius.

VI. Optimal Design Selection in the (l, b) Design Space

Figure 2 shows the parametric design plane, with l on the abscissa and b on the ordinate. The plot displays specific torque isolines in blue, and efficiency isolines in red, all expressed as functions of the design parameters l and b. It should be noted that this plot corresponds to a maximum disk volume of $V = \pi R_o^2 L_{st}$, with $R_o = 0.06$ m and $L_{st} = 0.02$ m.

The design of electric propulsion systems for UAV inherently involves trade-offs between multiple competing objectives, particularly between specific torque and efficiency. Specific torque is directly related to power density and to the total mass of the propulsion system, whereas efficiency has a direct impact on flight endurance, thermal operating conditions, and UAV reliability [22], [23]. Due to these competing effects, the simultaneous maximization of both metrics naturally leads to a multiobjective optimization problem. In this context, there is no single unique optimal solution, but rather a set of non-dominated solutions known as the Pareto front [9], [10]. From this Pareto-optimal set, an additional decision criterion must be introduced to select a final design that represents the best compromise between specific torque and efficiency, in accordance with the UAV mission requirements. In this work, three selection strategies widely adopted in the multiobjective optimization and engineering decision-making literature are compared: (i) weighted distance to the ideal point, (ii) weighted product aggregation, and (iii) knee-point selection [11], [24], [25]. To this end, a normalized design space $(\tilde{T}_{spec}, \tilde{\eta})$ is first constructed, where $\tilde{T}_{spec}$ denotes the normalized specific torque and $\tilde{\eta}$ denotes the normalized efficiency, defined as

$$\tilde{T}_{spec} = \frac{T_{spec}(l,b) - \min(T_{spec})}{\max(T_{spec}) - \min(T_{spec})} \tag{28}$$

$$\tilde{\eta} = \frac{\eta(l,b) - \min(\eta)}{\max(\eta) - \min(\eta)} \tag{29}$$

A. Weighted Distance to the Ideal Point

The ideal (or utopian) point is defined as the hypothetical solution that simultaneously maximizes all considered objectives. In the normalized objective space, this point corresponds to $(\tilde{T}_{spec}, \tilde{\eta}) = (1, 1)$ [24]. A compromise design can be selected by minimizing the weighted Euclidean distance to this point,

$$d = \sqrt{w_T(1 - \tilde{T}_{spec})^2 + w_\eta(1 - \tilde{\eta})^2}, \tag{30}$$

where w_T and w_η are non-negative weighting factors reflecting the relative importance of each objective [24], [25]. This approach allows efficiency to be explicitly prioritized ($w_\eta > w_T$) in order to maximize flight endurance and limit thermal dissipation, which are critical aspects in battery-powered aerial platforms [22], [23].

B. Weighted Product Aggregation

As an alternative to the distance-to-ideal criterion, a weighted product of the normalized objectives is employed,

$$J = \tilde{T}_{spec}^{\alpha} \tilde{\eta}^{\beta}, \tag{31}$$

where α and β are positive exponents controlling the relative priority of each metric. This approach is rooted in multiattribute utility theory and has been extensively studied as a robust scalar aggregation method for multiobjective decision problems [25], [26]. A relevant characteristic of the product method is that it strongly penalizes solutions exhibiting poor performance in any one of the objectives, even if the other objective attains high values [25]. This property enables the rejection of configurations with high torque density but unacceptably low efficiency, or vice versa, thereby favoring balanced and operationally robust designs under varying load and operating conditions [23].

C. Knee-Point Selection

The knee point of the Pareto front corresponds to a solution for which a marginal improvement in one objective results in a disproportionate degradation of the other [11]. From a geometric perspective, the knee point can be identified as the Pareto-optimal solution that maximizes the distance to the line connecting the extreme points of the front [11]. This criterion does not require the explicit definition of weights or preferences and provides an objective reference for the intrinsic trade-off of the design problem. The knee point therefore yields a balanced compromise between efficiency and specific torque, particularly useful when detailed mission priorities are not available.

D. Comparison of Criteria and Selection Strategy

The three considered methods provide complementary perspectives for final design selection. The distance-to-ideal criterion enables the explicit incorporation of mission priorities and is particularly suitable for UAV targeting maximum endurance and thermal margin [22]. The weighted product method emphasizes robustness of the compromise and reduces sensitivity to arbitrary scaling of the performance metrics [25]. The knee-point criterion provides a neutral solution independent of external preferences [11]. Figure 3 shows the Pareto front, highlighted in yellow, together with the designs selected using the aforementioned criteria. As can be observed, all methods select designs belonging to the Pareto-optimal set. Figure 4 presents the Pareto-optimal designs and the solutions selected by the three methods projected onto the (l, b) design plane. The region corresponding to Pareto-optimal designs within the selected design space can be clearly identified. It is also observed that the selection criteria are robust, as no substantial differences arise among the selected solutions. In this work, the final UAV electric propulsion design is selected from the Pareto-optimal set using the knee-point criterion, while the solutions obtained through weighted product aggregation and weighted distance to the ideal point are presented as comparative references to assess the robustness and sensitivity of the adopted decision.

VII. FEA Validation

In this section, we perform a 2D FEA of the design selected and compare it with the results obtained from the analytical model. For the FEA, we use FEMM [27]. Although three-dimensional effects are neglected, the 2D FEA is sufficient to validate the electromagnetic quantities of interest, which are dominated by radial flux components in the proposed

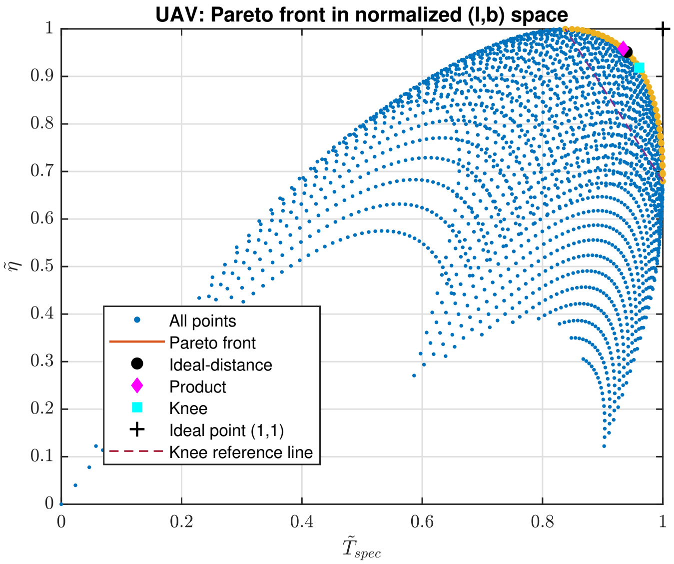


Fig. 3: Pareto front as function of $\tilde{T}_{spec}$ and $\tilde{\eta}$

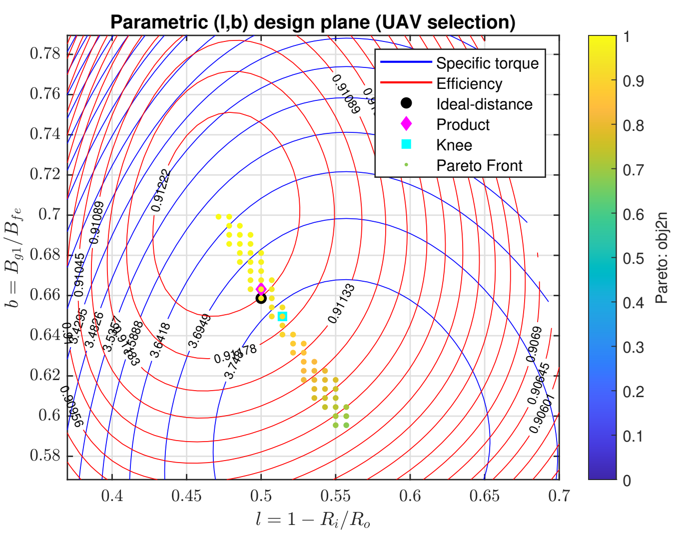
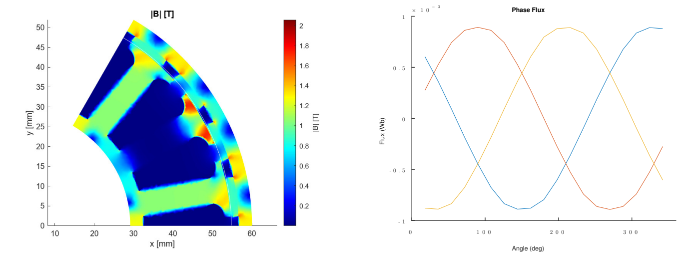


Fig. 4: Selected designs projected onto the (l, b) design plane.

outer-rotor topology. Figure 5(a) and (b) show the magnetic flux density distribution in the stator and rotor under no-load conditions and the linkage flux, respectively. As can be observed, the maximum magnetic flux density in both the rotor and stator remains below the specified design limits. In addition, no excessive local flux concentration is observed.

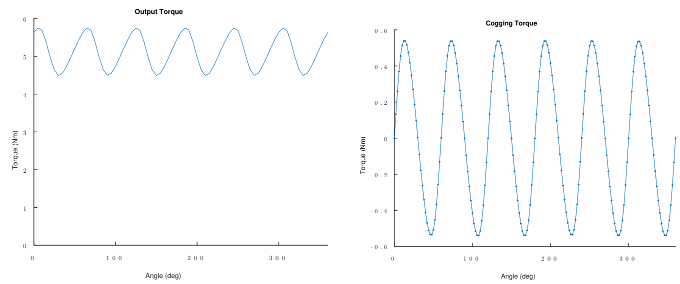
The electromagnetic output torque and the cogging torque are shown in Figure 6(a) and (b), respectively. The peak cogging torque is approximately 10% of the average output torque, which is considered acceptable for this class of machines. The average output torque closely matches the value predicted by the analytical model, while the observed torque ripple is mainly attributed to cogging torque effects. Figure 7 presents the efficiency map over the entire operating range. As expected for eVTOL propulsion applications, the maximum efficiency is achieved in the region of high torque and high rotational speed.



(a) Magnetic flux density distribution under no-load conditions.

(b) Phase flux linkage waveform.

Fig. 5: FEA results: (a) magnetic flux density distribution and (b) phase flux linkage waveform.



(a) Output torque.

(b) Cogging torque.

Fig. 6: FEA results: (a) electromagnetic output torque and (b) cogging torque.

Table I summarizes the physical parameters of the selected machine, while Table II compares the values obtained from the analytical model with those derived from FEA simulations.

VIII. Conclusions

A systematic method to identify the Pareto-optimal design of an outer-rotor surface-mounted permanent magnet syn-

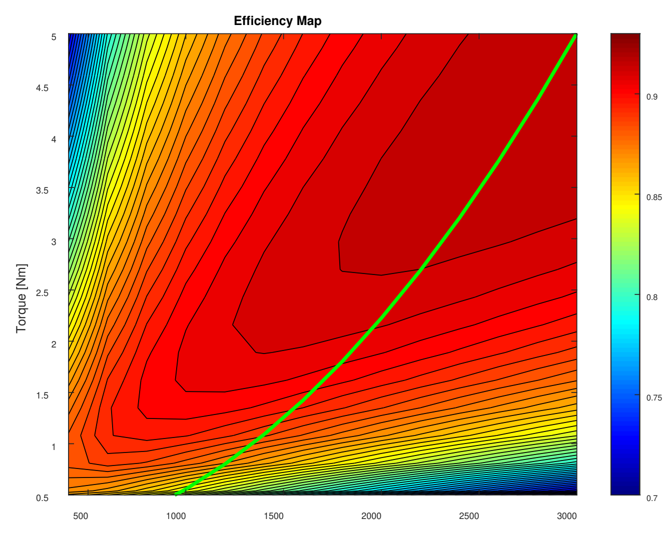


Fig. 7: Efficiency map obtained by FEA.

TABLE I: Selected PMSM parameters

Parameter	Value	Unit
Magnets	24	
Slots	18	
Output radius	0.06	m
Airgap radius	0.055	m
Airgap length	0.0008	m
Magnets size	0.02 x 0.01 x 0.0012	(L x W x T) m
α_m	0.69π	rad
Slot opening	0.0048	m
Slot length	0.018	m
Leg width	0.0063	m
Backiron thickness	0.0032	m

TABLE II: PMSM Key Performance Indicators from Analytical Model and FEA

Value	Model	FEA	Unit	Error (%)
Phase current	333	333	A	
k_T	0.0084	0.0089	Nm/A	-5.6
Torque	4.85	5.12	Nm	-5.6
λ	0.81	0.89	mWb	9.87
B_g (max)	0.84	0.93	T	-11
Specific torque	3.77	4.33	Nm/kg	-14.6
Efficiency	91.1	93.4	%	-0.8
Total mass	1.28	1.15	kg	10.2

chronous machine (SM-PMSM) within a predefined design space has been presented, specifically targeting eVTOL UAV propulsion applications. To this end, a parametrized analytical design model based on the (l, b) design plane was developed, where the introduction of the parameter l is required to properly capture the characteristics of this machine geometry. In accordance with the application requirements, specific torque and efficiency were selected as the key performance indicators (KPIs).

To select the most suitable design among all feasible solutions within the design plane, three multiobjective selection techniques were analyzed and compared: weighted distance to the ideal point, weighted product aggregation, and knee-point selection. The results show that all three methods yield similar near-optimal solutions when equal weighting factors ($\alpha = 1$, $\beta = 1$) are employed. Based on this analysis, the final design is selected from the Pareto-optimal set using the knee-point criterion.

Finally, the selected SM-PMSM design was evaluated using finite element analysis (FEA) performed in FEMM, and the results were compared with those obtained from the proposed analytical model. The comparison demonstrates that the analytical model predicts the selected KPIs with good accuracy, validating its suitability for early-stage design and optimization of outer-rotor SM-PMSMs for eVTOL UAV applications.

REFERENCES

[1] T. F. Tallerico, "Nasa reference motor designs for electric vertical takeoff and landing vehicles," in *AIAA/IEEE Electric Aircraft Technologies Symposium*, NASA, 2021.

[2] A. Bacchini and E. Cestino, "Electric vtol configurations comparison," *Aerospace*, vol. 6, no. 3, p. 26, 2019.

[3] Q. He, H. Wang, and Q. Li, "Design optimization and performance analysis of high torque density and efficiency motors for evtol applications," *IEEE Access*, 2025.

[4] J. Zong, B. Zhu, Z. Hou, X. Yang, and J. Zhai, "Evaluation and comparison of hybrid wing vtol uav with four different electric propulsion systems," *Aerospace*, vol. 8, no. 9, p. 256, 2021.

[5] J. Z. Bird, "A review of electric aircraft drivetrain motor technology," *IEEE Transactions on Magnetics*, 2021.

[6] C. Doppler, F. Holzapfel, and M. K. Scharrer, "Requirements and design of powertrains for evtols," *e+i Elektrotechnik und Informationstechnik*, 2024.

[7] W. J. Fredericks, M. D. Moore, and C. R. Busan, "Benefits of hybrid-electric propulsion to achieve 4x cruise efficiency for a vtol uav," in *AIAA International Powered Lift Conference*, AIAA, 2013.

[8] S. Grimshaw, J. Taylor, P. Nathen, A. Strohmayer, and R. Miller, "Architectural performance assessment of an electric vertical take-off and landing (e-vtol) aircraft based on a ducted vectored thrust concept." https://www.lilium.com/, 2021. Lilium GmbH, Claude Dornier.

[9] V. Pareto, "Cours d'économie politique," 1896. Lausanne.

[10] K. Deb, *Multi-Objective Optimization Using Evolutionary Algorithms*. John Wiley & Sons, 2001.

[11] K. Deb and S. Gupta, "Understanding knee points in bicriteria problems and their implications as preferred solution principles," *Engineering Optimization*, vol. 35, no. 2, pp. 117–148, 2003.

[12] J. G. Leishman, *Principles of Helicopter Aerodynamics*. Cambridge University Press, 2nd ed., 2006.

[13] J. Staszak, "Ac distributed winding with low higher spatial harmonics content in mmf distribution," *Energies*, vol. 16, no. 14, 2023.

[14] P. Rasmussen, J. Davies, T. Miller, M. McGelp, and M. Olaru, "Analytical and numerical computation of air-gap magnetic fields in brushless motors with surface permanent magnets," *IEEE Transactions on Industry Applications*, vol. 36, no. 6, pp. 1547–1554, 2000.

[15] D. Hanselman, "Brushless permanent magnet motor design," 01 2003.

[16] L. Zhu, S. Z. Jiang, Z. Q. Zhu, and C. C. Chan, "Analytical methods for minimizing cogging torque in permanent-magnet machines," *IEEE Transactions on Magnetics*, vol. 45, no. 4, pp. 2023–2031, 2009.

[17] B. Boazzo, G. Pellegrino, and A. Vagati, "Multipolar spm machines for direct-drive application: A general design approach," *IEEE Transactions on Industry Applications*, vol. 50, no. 1, pp. 327–338, 2014.

[18] C. Lu, S. Ferrari, and G. Pellegrino, "Two design procedures for pm synchronous machines for electric powertrains," *IEEE Transactions on Transportation Electrification*, vol. 3, no. 1, pp. 98–107, 2017.

[19] C. Lu and G. Pellegrino, "A simple design method for surface-mounted pm machines for traction application," in *2016 IEEE Energy Conversion Congress and Exposition (ECCE)*, pp. 1–7, 2016.

[20] D. Numan, B. Jorge M., A. Rodrigo, and C. Esteban, "A computer-simulated environment for modeling and dynamic- behavior-analysis of special brushless motors for mechatronic mobile robotics systems," *IFAC-PapersOnLine*, vol. 49, no. 29, pp. 12–17, 2016. 17th IFAC Conference on International Stability, Technology and Culture TECIS 2016.

[21] R. Alcoberro, J. Bauer, and N. M. Durakbasa, "Direct-drive integrated motor/controller with iot and gis management for direct sowing and high efficiency food production," in *Digitizing Production Systems* (N. M. Durakbasa and M. G. Gençyılmaz, eds.), (Cham), pp. 271–280, Springer International Publishing, 2022.

[22] J. G. Leishman, *Principles of Helicopter Aerodynamics*. Cambridge University Press, 2006.

[23] M. McCrink, J. Gregory, and N. Alley, "Electric propulsion system considerations for small unmanned aerial vehicles," *IEEE Aerospace and Electronic Systems Magazine*, vol. 32, no. 6, pp. 28–38, 2017.

[24] K. Miettinen, *Nonlinear Multiobjective Optimization*. Kluwer Academic Publishers, 1999.

[25] R. T. Marler and J. S. Arora, "Survey of multi-objective optimization methods for engineering," *Structural and Multidisciplinary Optimization*, vol. 26, pp. 369–395, 2004.

[26] R. L. Keeney and H. Raiffa, *Decisions with Multiple Objectives: Preferences and Value Tradeoffs*. John Wiley & Sons, 1976.

[27] D. Meeker, *Finite Element Method Magnetics (FEMM)*, 2015. Version 4.2, http://www.femm.info.